

Handbook on the Use of 5x3 in Collaborative Statistical Design and Reporting

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INTRODUCTION

This handbook outlines steps in collaborative design and reporting of studies using five-region testing at multiple alpha levels. The accompanying document *5x3: A Practical Approach to Encouraging Thoughtful Statistical Analyses* justifies this approach to statistical analysis and describes the web-based visualization tool 5x3 that supports it.

As in standard statistical analyses, most of the effort comes in the planning stage and before data collection.

Traditional study designs require stakeholder differences to be resolved into consensus opinion. A virtue of five-region testing at multiple alpha levels is that it accommodates differences in anticipated effect sizes, notions of error costs etc.

This handbook does not provide guidance on how to conduct the group discussions required in the collaborative process. For this, refer to the extensive literature on stakeholder engagement in research (e.g. Boaz et al, 2018) and on formal processes such as the Delphi method to elicit opinions and work toward shared understandings.

WHO ARE YOUR STAKEHOLDERS AND WHY INVOLVE THEM IN YOUR RESEARCH?

The start point for you as a researcher is identifying who has potential interest in your research outcomes (and of course, who might fund your research). These people and/or organisations are the stakeholders in your research and should be seen as your collaborators. They can be donors, practitioners, or members of community organisations, trade groups, professional associations and so on.

For example, in research concerning elite sporting performance, stakeholders include professional athletes, coaches, team support members and possibly suppliers of equipment. If training injuries are an issue in the sport, the sports' administrators may be influential stakeholders, and if the research has direct or indirect academic funding, university staff may be stakeholders.

You don't approach stakeholders with a fully formed research plan. Huzzard (2021) says stakeholders should

- inform the research question and study design;
- facilitate the research implementation—for example, through access to subjects; and
- promote or make use of results.

Obviously, the degree of a stakeholder's willingness to be involved will depend on their expected payoff from research outcomes, as well as on the nature of the research (Hendricks, 2021). A study investigating the potential benefit of a change to a training routine for elite professional football teams might attract a stakeholder group consisting of a motivated coach, three senior athletes, a team physiotherapist, and a team psychologist. Typically, the sport scientist leading the research will have previously worked with at least some of these volunteers.

Getting the attention of stakeholders is not always easy. They are not a homogeneous group and can have strong differences in aims and opinions. Their involvement might seem at times to get in the way of your research rather than help it. Yet what is that research for, if not for them?

THE 8-STEP GUIDE FOR COLLABORATIVE STATISTICAL DESIGN AND ANALYSIS

The 8 steps start after you have organized a group of stakeholders who have agreed to participate in the collaborative research effort, and you have sorted out logistical details such as meeting places and times.

The steps are then divided into a block of five, taken before data are collected (part A), and three analysis and reporting steps (part B). See Figs 1 and 2.

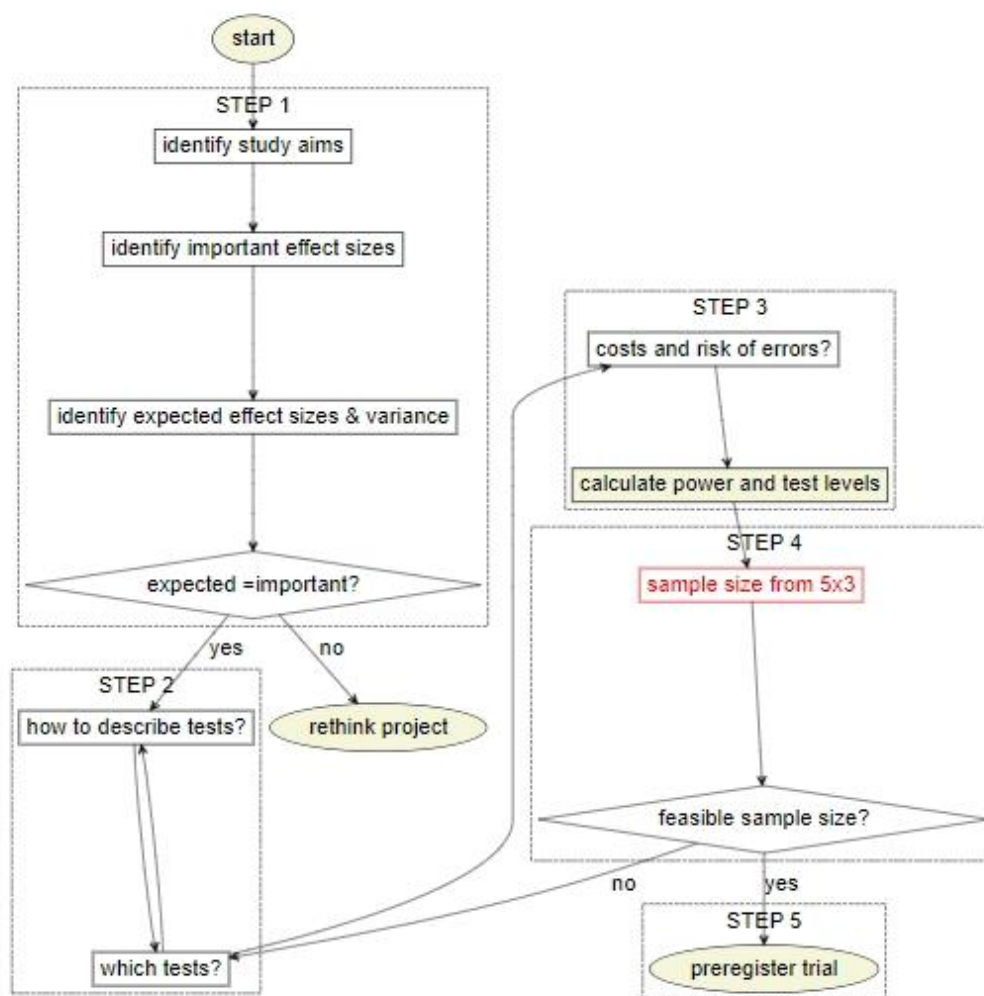


Fig. 1 The five steps in PART A: Statistical design and registration of the study. All activities require collaboration with stakeholders except for the beige coloured items.

PART A: Statistical design and registration of the study

Fig. 1 summarizes the steps in Part A.

STEP 1. Tentatively propose effect sizes that matter and likely effect size

Discuss with your stakeholders their individual perceptions of the study aims and what “success” looks like. From this, collaboratively determine what effect sizes are practically meaningful; if consensus cannot be reached, allow more than one option for the moment.

Discuss what the anticipated effect size is and its likely variance in the population of interest. The lead researcher should present evidence from previous studies as appropriate. Do not aim for consensus but do discuss whether to proceed with the trial if some stakeholders believe the intervention will not be a “success”.

STEP 2. Determine what to test and the terminology to describe test and findings

Determining what to test and deciding on the terminology to use are intertwined, since the description of the five-region tests must be meaningful to the stakeholders before they can express which are relevant to them. For instance, everyone might anticipate a large positive effect size, in which case only the test for superiority might be relevant, and this terminology might be acceptable to all. If opinion is divided about likely effectiveness, findings that are compatible with saying “intervention helps” (superiority) or “no improvement and might even be worse” (non-superiority) may be relevant. Some stakeholders may be open minded as to whether the intervention will or will not be an improvement, while a doubter—the physiotherapist in our case—might specifically want their prediction that “the intervention is going to be worse than current practice” to be put to the test.

STEP 3. Determine attitudes to errors to set alpha levels and power levels

Discuss possible consequences if the intervention is “successful” but the findings suggest it is not, and vice versa. What about if the intervention is no worse than current practice, but findings suggest it is? How do stakeholders rate the chance of the intervention bringing meaningful improvement?

Try to agree on terminology to describe the control of Type I error rates provided by different test levels, such as what would be considered “strong”, “medium” and “weak” control. Consensus as to the “correct” terminology is not essential provided stakeholders and the lead researcher all know what each other means.

From this discussion, use expected total error costs to either formally or informally identify one or more pairs of Type I and Type II error rates for the nominated tests using standard procedures to minimize expected total error costs. Guidance on this is provided by Mudge et al (2012) and Maier & Lakens & Meier (2022), while Aisbett (2024) addresses costs when testing at multiple alpha levels.

Note that the views of a stakeholder who predicts that the intervention will bring improvement only affects power and alpha level settings for tests for superiority and, possibly, non-inferiority, while the doubter might only influence settings for inferiority. Tests at more than one alpha level can accommodate different stakeholders’ beliefs in the costs of errors.

STEP 4. Agree on a feasible sample size, and possibly adjust study

Use the possibly diverse views of the stakeholders collected in the earlier steps to complete the tables in the *Sample size* tab of the 5x3 tool. Use the tool to estimate the maximum sample size that delivers desired power at each test and each nominated alpha level, given the margins of material significance and effect sizes anticipated by different stakeholders. Identify resource constraints that might limit this sample size.

If needed, adjust power levels, test levels, and margins using the visual depiction provided by the tool to help stakeholders understand the implications of their preferences.

For example, the desire for a high level of evidence for improved performance might be replaced by standard level evidence if, at the same time, there is high-level evidence for the new intervention not having inferior performance. Other examples illustrating the use of the 5x3 tool are given in the accompanying 5x3 user manual.

Adjust the suite of tests if necessary to determine a final set of tests at alpha levels that have acceptable power for a feasible sample size.

STEP 5. Pre-register the trial

Preregistration of research proposals is becoming standard practice in experimental disciplines to improve transparency and to ensure results of “unsuccessful” trials are not buried (e.g., Caldwell et al, 2020, in the case of sport science). Preregistration typically involves entering the research hypotheses and statistical design on a publicly available repository such as OSF <https://help.osf.io/article/158-create-a-preregistration>. This is done before data are collected. Typically, pre-registration requires nominating research as exploratory or confirmatory; research that accommodates different stakeholder views in an analysis involving 5-region testing at multiple alpha levels is exploratory.

PART B. Analysis and reporting of the study

Part B assumes that your data have been collected and pre-processed and that summary statistics of effect sizes and variances or standard errors have been computed. The 3 steps to be taken in the collaborative process are then summarized in Fig. 2.

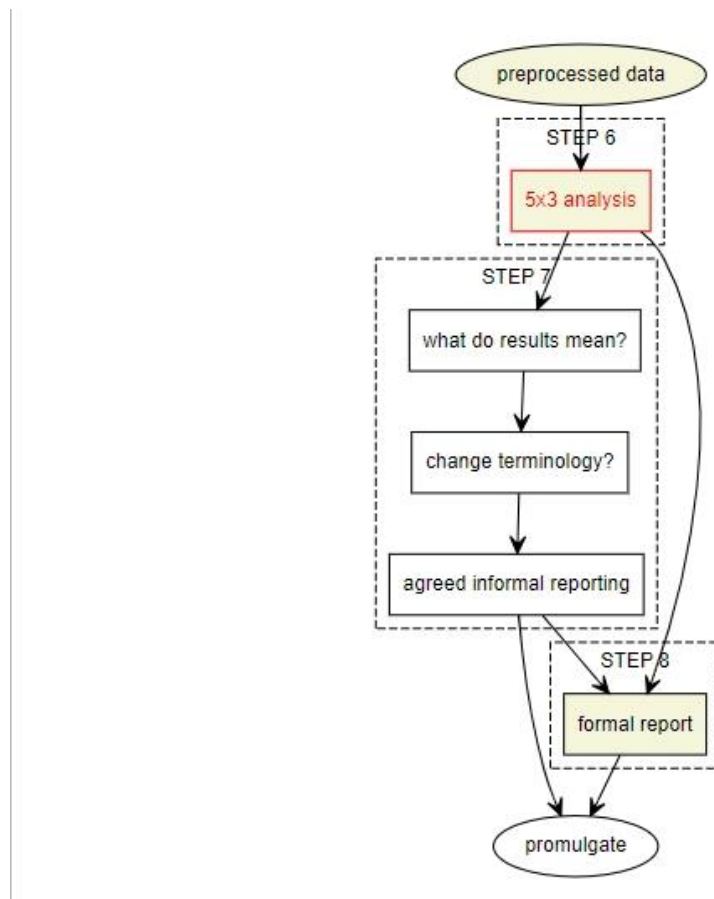


Fig. 2. The 3 steps in Part B, analysis and reporting of the study, shown between the initial pre-processing of the data and final task of promulgating findings. Beige items are performed by the lead research team, all others are collaborative with stakeholders.

STEP 6. Preliminary analysis

Prepare preliminary report of findings using the graphical representations of the summary data provided by 5x3. These consist of charts of rejection regions and multi-level confidence intervals obtained from the tests determined in Part A and are described in the accompanying user manual. Report preliminary findings in a way that will be meaningful to stakeholders.

STEP 7. Informal reporting, with lessons

Discuss findings with stakeholders. Discuss what conclusions stakeholders would draw, and how they would report them. There is no need to use previously agreed terminology, so change this if stakeholders prefer. Is the consensus that the intervention was useful? What is the way forward if it is? If it is not, what could be learned from how the study was designed and conducted?

STEP 8. Formal reporting

Report the study in scientific papers describing findings in formally correct statistical terminology, for example, that the hypothesis that the effect is less than U was rejected at an alpha level of 0.05 but was not rejected at alpha level 0.01. Also report stakeholders' interpretations of the findings.

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